***Brock Biology of Microorganisms, 15e* (Madigan et al.)**
Chapter 27 Adaptive Immunity: Highly Specific Host Defenses

27.1 Multiple Choice Questions

1) T cells recognize antigens with their
A) antibodies.
B) leukocidins.
C) M proteins.
D) T cell receptors.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

2) Antibody-mediated immunity is particularly effective against
A) extracellular pathogens.
B) intracellular pathogens.
C) both extra- and intra-cellular pathogens.
D) None of the answers are correct.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

3) ________ is the acquired inability to mount an adaptive immune response against self.
A) Memory
B) Specificity
C) Tolerance
D) Immunogenicity
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

4) Intrinsic properties of immunogens include
A) appropriate physical form.
B) molecular size.
C) sufficient molecular complexity.
D) appropriate physical form, molecular size, and molecular complexity.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

5) The part of the antigen recognized by the antibody or TCR is called the
A) epitope.
B) antigen-binding site.
C) antigenic complex.
D) light chain.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

6) A cross-reaction is an interaction between an antibody or TCR and
A) a heterologous antigen.
B) a homologous antigen.
C) either a heterologous or homologous antigen.
D) None of these are correct.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

7) Which of the following are found on the surfaces of all nucleated cells?
A) Class I MHC proteins
B) Class II MHC proteins
C) G proteins
D) M proteins
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.5

8) T_H cells express a ________ protein coreceptor.
A) CD4
B) CD8
C) CD12
D) T cell receptin
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.5

9) Natural killer cells like Tc cells use ________ and granzymes to kill their targets without prior exposure or contact with the foreign cells.
A) coagulants
B) M protein
C) perforin
D) interferon gamma
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.5

10) Which of the following enzymes are NOT secreted by TH1 to activate macrophages?
A) granulocyte-monocyte colony stimulating factor
B) interferon gamma
C) TNF-α
D) fibrin
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.8

11) Antibodies are found in
A) milk.
B) mucosal secretions.
C) serum.
D) milk, mucosal secretions, and serum.
Answer: D
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 27.3

12) Which of the following is NOT an immunoglobin?
A) IgA
B) IgC
C) IgD
D) IgG
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.3

13) The most common circulating antibody, comprising about 80% of the serum immunoglobin, is
A) IgA.
B) IgC.
C) IgD.
D) IgG.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.4

14) The measurable strength of the binding of an antibody to antigen is called
A) binding affinity.
B) epitope.
C) tolerance.
D) virulence.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 25.7

15) Which of the following is the name of the heavy chain constant domain of an IgD antibody?
A) beta
B) delta
C) zeta
D) rho
Answer: B
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.4

16) Dimers of IgA are present in
A) breast milk colostrum.
B) saliva.
C) tears.
D) breast milk colostrum, saliva, and tears.
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.3

17) Antigen exposure is necessary to
A) induce genetic hypermutation in B cells with productive antibody genes.
B) stimulate B cells to differentiate to plasma cell.
C) stimulate B cells to produce soluble antibodies.
D) stimulate B cell differentiation, antibody production, and genetic hypermutation of antibody genes.
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.4

18) In class switching during the secondary antibody response, the most common antibody switch is from
A) IgA to IgD.
B) IgD to IgE.
C) IgM to IgG.
D) None of the answers are correct.
Answer: C
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.3

19) Which of the following is a probable lifespan of memory B cells?
A) 5 minutes
B) 5 hours
C) 5 days
D) 5 years
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

20) Antigen-presenting cells present antigens to
A) B lymphocytes.
B) T lymphocytes.
C) dendritic cells.
D) neutrophils.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

21) Substances that induce an immune response are known as
A) immunogens.
B) antigens.
C) immunoglobins.
D) antigen-presenting cells.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

22) Which of the following does NOT influence immunogenicity?
A) the routes of administration
B) foreign nature of the immunogen to the host
C) solubility
D) All of these influence immunogenicity.
Answer: D
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 27.2

23) The antigen that induces an antibody or TCR is called the
A) homologous antigen.
B) heterologous antigen.
C) cross-reactive antigen.
D) immunogenic antigen.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

24) Which antibodies bind complement?
A) IgA
B) IgB
C) IgM
D) IgC
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.3

25) Proteins derived from infecting viruses are taken up and digested in the cytoplasm in a structure called the
A) phagosome.
B) liposome.
C) proteasome.
D) nucleosome.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.5

26) T cell-target cell interactions induce specialized Tc cells to produce ________ that kill virus-infected target cells.
A) perforins
B) granulozymes
C) leukocidins
D) pyogens
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.8

27) TC cells express a(n) ________ protein co-receptor.
A) CD4
B) CD8
C) IgM
D) IgG
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.5

28) Which of the following is associated with mucosa-associated lymph tissue that produces IgA?
A) breast milk
B) red blood cells
C) blood serum
D) small intestine
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.3

29) A serum containing antigen-specific antibodies is called an
A) antiserum.
B) antitoxin.
C) antiantigen.
D) anticoagulase.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

30) Which of the treatments listed below would be most effective for a patient with a genetic defect that causes severe combined immunodeficiency (SCID), a condition in which there is a severe deficiency in B and T lymphocytes?
A) intravenous antibiotics
B) bone marrow transplant
C) multiple immunizations
D) repeated doses of antisera
Answer: B
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 26.4

31) Which is an example of acquisition of natural passive immunity?
A) a fetus protected from disease by its mother′s antibodies
B) a person who received his or her yearly influenza vaccine
C) a person who acquired the chickenpox
D) a person who received tetanus antiserum after stepping on a rusty nail
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

32) Which of these is often used as a prophylactic measure to protect a person against future attack by a pathogen?
A) antiserum injection
B) artificial passive immunity
C) vaccination
D) None of these are correct.
Answer: C
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 27.2

33) Type I hypersensitivity is caused by the release of vasoactive products from mast cells coated with
A) IgA.
B) IgE.
C) IgG.
D) IgM.
Answer: B
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.9

34) In Hashimotos′s disease, auto-antibodies are made against ________, a product of the thyroid gland.
A) bacteria
B) hemoglobin
C) thyroglobulin
D) viruses
Answer: C
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.9

35) Systemic lupus erythematosus (SLE) is an example of a disease caused by type ________ hypersensitivity.
A) I
B) II
C) III
D) IV
Answer: C
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.9

36) Which of the following diseases is/are associated with organ-specific autoimmune disease?
A) autoimmune hypothyroidism
B) juvenile (type I) diabetes
C) systemic lupus erythematosus (SLE)
D) autoimmune hypothyroidism and juvenile (type I) diabetes
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.9

37) The life-threatening condition that may develop during a type I hypersensitivity reaction is called
A) anaphylaxis.
B) septic shock.
C) stroke.
D) toxic shock.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.9

38) All the following can cause contact dermatitis EXCEPT
A) jewelry.
B) latex gloves.
C) poison ivy.
D) superantigens.
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.9

39) After the first exposure to an antigen, a ________ stimulates growth and multiplication of antigen-reaction cells.
A) secondary innate immune response
B) primary adaptive immune response
C) phagocytic immune response
D) hyperactive cytotoxic response
Answer: B
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

40) A diagnostic test for *M. tuberculosis* exposure
A) utilizes an inflammatory reaction that occurs if there has been previous exposure to *M. tuberculosis* antigens.
B) relies on previous tuberculin vaccination.
C) uses an antiserum to detect antibodies in the blood that are only formed if there has been previous exposure to *M. tuberculosis* antigens.
D) relies on an innate immune response that occurs only if there has been previous exposure to *M. tuberculosis* antigens.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.9

41) The transfer of antibodies through the placenta from mother to fetus is an example of
A) natural passive immunity.
B) innate immunity.
C) natural active immunity.
D) artificial active immunity.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

42) The purposeful artificial stimulation of active immunity to a particular infectious disease is known as
A) immunodeficiency.
B) hypersensitivity.
C) immunization.
D) opsonization.
Answer: C
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

43) What are the primary chemical mediators released from mast cells during a type I hypersensitivity reaction?
A) antibodies
B) interleukin I and tumor necrosis factor
C) histamine and serotonin
D) cytokines
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.3

44) Which of the following is NOT an autoimmune disease?
A) systemic lupus erythematosus
B) anaphylaxis
C) type 1 diabetes
D) multiple sclerosis
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.9

45) Superantigens produce a harmful immune response because
A) they activate too many T cells, causing excessive inflammation and cell damage.
B) they initiate a type I allergic response resulting in excessive inflammation.
C) too many phagocytes are activated and destroy host tissue.
D) a strong immune response is necessary to protect the host from the pathogen.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.10

46) The C domain of antigen-binding proteins functions to
A) attach the antigen-binding domain to the cytoplasmic membrane.
B) bind to the cytoplasmic membrane of foreign cells.
C) complex multiple antigen-binding proteins together.
D) bind to epitopes.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.7

47) The main bonds present in the C domain of antigen-binding proteins are
A) hydrogen bonds.
B) intrachain disulfide bonds.
C) ionic bonds.
D) van der Waals forces.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.7

48) Which of the following is/are MHC class I genes?
A) HLA-A
B) HLA-B
C) HLA-C
D) HLA-A, HLA-B, and HLA-C
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.6

49) An individual usually expresses ________ structurally distinct allele(s) that encode class I MHC proteins.
A) one
B) two
C) four
D) six
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.6

50) Immunoglobulin G consists of four polypeptides, ________ heavy chains, and ________ light chains.
A) one / three
B) two / two
C) three / one
D) four / four
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.3

51) There are ________ different complementarity-determining regions in each V domain of an immunoglobulin (meaning ________ total in the V region).
A) one / two
B) two / four
C) three / six
D) four / eight
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.4

52) An epitope is ________ amino acids long.
A) 3-5
B) 10-15
C) 50-60
D) 700
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.4

53) Clonal anergy is a state of
A) antibody production.
B) high energy.
C) unresponsiveness.
D) high energy and antibody production.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

54) Which of the following contribute to the limitless diversity generated from a relatively fixed number of immunoglobulin genes?
A) hypermutation only
B) random heavy and light chain reassortment only
C) somatic recombination only
D) hypermutation, random heavy and light chain reassortment, and somatic recombination
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.4

55) Assuming that each heavy and light chain has an equal chance to be expressed in each cell, approximately how many possible antibodies can be expressed?
A) 192
B) 1,920
C) 19,200
D) 3,360,000
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.4

56) Somatic hypermutation occurs only in the ________ regions of rearranged heavy and light chains.
A) D
B) HV
C) J
D) V
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.4

57) The two-stage thymic selection process for selecting self-tolerant, antigen-reactive T cells results in
A) allelic exclusion.
B) clonal deletion.
C) complementarity-determining regions.
D) somatic hypermutation.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

58) More than ________% of all T cell precursors that enter the thymus do NOT survive the selection process.
A) 15
B) 25
C) 45
D) 95
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

59) Which of the following is involved in B cell selection and tolerance?
A) clonal anergy
B) clonal deletion
C) clonal anergy and deletion
D) allelic exclusion
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

60) The complete second signal for B cell activation requires which of the following interactions?
A) Interleukin-4 release
B) cytokine-cytokine receptor interaction
C) T_H-B cell interaction
D) All of these are necessary.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

61) Which of the following are soluble mediators and activators for T lymphocytes and B cells?
A) IL-2
B) IL-4
C) IL-5
D) IL-2, IL-4, and IL-5
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.8

62) Which of the following is secreted by Th1 cells to stimulate macrophage differentiation and activation?
A) granulocyte-monocyte colony stimulating factor
B) interferon gamma
C) tumor necrosis factor-alpha
D) granulocyte-monocyte colony stimulating factor, interferon gamma, and tumor necrosis factor-alpha
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.8

63) Which of the following cytokines is produced by macrophages and dendritic cells and acts on natural killer cells and naïve T cells?
A) IL-1β
B) IL-6
C) IL-12
D) TNF-α
Answer: C
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.8

64) Chemokines produced by activated macrophages include
A) CXCL8.
B) TNF-α.
C) interleukin-4.
D) interleukin-5.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.8

65) The chemokine CCL2 is produced by
A) Th2 cells.
B) dendritic cells.
C) basophils.
D) macrophages.
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.8

66) The ________ is a highly conserved amino acid sequence found in immunoglobulin, TCR, and MHC proteins.
A) heavy chain
B) V domain
C) C domain
D) light chain
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.7

67) The major histocompatibility complex in humans is also called the
A) human leukocyte antigen (HLA).
B) T cell receptor.
C) toll-like receptor complex.
D) T cell receptor or toll-like receptor.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.5

68) The ________ and ________ domains of the class II MHC protein interact to form a peptide-binding site.
A) B / T
B) light / heavy
C) C / V
D) α1 / β1
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.5

69) The occurrence of multiple alleles at a specific locus is called
A) signal transduction.
B) the variable domain.
C) polymorphism.
D) allelic exclusion.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.6

70) The peptides bound by a single MHC protein share a common
A) C domain.
B) peptide motif.
C) V domain.
D) nucleotide sequence.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.6

71) The invariant amino acids in each peptide motif bound by MHC proteins are called
A) anchor residues.
B) MHC peptides.
C) antigen-binding immunoglobulins.
D) TCR peptides.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.6

72) Amino acid variability is especially apparent in several ________ in the variable domains of different immunoglobulins.
A) heavy chains
B) complementarity-determining regions (CDR)
C) peptide motifs
D) epitopes
Answer: B
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 27.4

73) The antigen-binding site of an antibody accommodates a small portion of the antigen called a(n)
A) lipopolysaccharide.
B) epitope.
C) peptide motif.
D) pathogen associated molecular pattern (PAMP).
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.4

74) ________ ensures that each B cell produces only one immunoglobulin.
A) Self-activation
B) Allelic exclusion
C) MHC interaction
D) Clonal deletion
Answer: B
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 27.4

75) The mechanism by which antigen-reactive cells respond to foreign antigens while ignoring self-antigens is called clonal
A) selection.
B) deletion.
C) anergy.
D) expansion.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.9

76) In B cell selection and tolerance, if no signal is generated from ________ cells, the B cells remain unresponsive.
A) T_C
B) T_H
C) B7
D) CD28
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

77) Some activated B cells are transformed into ________ cells that secrete antibodies, and others remain as ________ cells.
A) plasma / memory B
B) lymphocytes / memory B
C) immunoglobulin / plasma
D) immunoglobulin / T_H
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

27.2 True/False Questions

1) Th2 cells produce a cytokine that promotes growth and activation of other T cells and activates macrophages.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.3

2) Th17 cells are important in the first stages of the adaptive immune response.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.8

3) The antigen-binding site of all antibodies form by interactions between the variable domains of the heavy and light chains.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.4

4) T cell receptors on a particular T cell can recognize more than two antigens.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

5) In the adaptive immune response, effective immunity cannot be detected for several days after the first contact with the pathogen.
Answer: TRUE
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 27.1

6) Tolerance is the acquired ability to make an adaptive immune response directed to self-antigens.
Answer: FALSE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

7) All antigens are immunogens.
Answer: FALSE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

8) Haptens CANNOT bind to or induce an immune response but CAN bind to antibodies.
Answer: TRUE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

9) A high-affinity antibody binds nonspecifically and loosely to an antigen.
Answer: FALSE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.9

10) TCRs recognize epitopes only after the immunogens have been partially degraded.
Answer: TRUE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

11) The inducing antigens that react with an antibody are called heterologous antigens.
Answer: FALSE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

12) HLAs are responsible for immune-mediated organ transplant rejection.
Answer: TRUE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.5

13) Class I MHC proteins are found ONLY on the surface of B lymphocytes, macrophages, and dendritic cells.
Answer: FALSE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.5

14) Self-reactive T cells are eliminated during the development of tolerance in the immune system.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

15) CD4 and CD8 proteins are used for *in vitro* tests as T cell markers to differentiate T_H cells from T_C cells.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.5

16) IgE is found in extremely small amounts in serum.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.4

17) T_H2 cells interact directly with the pathogens.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.3

18) An individual can receive injections of an antiserum or purified antibodies derived from an immune individual.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

19) Individuals with severe combined immunodeficiency syndrome (SCID) have a genetic defect that prevents proper formation and expression of immunoglobins and T cells and therefore do not have adaptive immunity.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.10

20) The use of antivenom to treat snakebite is an example of artificial passive immunity.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.2

21) Antihistamines are used to treat some allergic symptoms because they neutralize the histamine mediators that cause rapid dilation of blood vessels and contraction of smooth muscles that initiate the symptoms of systemic anaphylaxis.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.9

22) Symptoms of delayed-type hypersensitivity may appear several hours to days after secondary exposure to the eliciting antigen.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.9

23) Men are 20 times more likely to develop systemic lupus erythematosus (SLE) than women.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.9

24) Superantigens activate more T cells than a normal immune response.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.10

25) *Staphylococcus aureus* and *Streptococcus pyogenes* produce exotoxins known as superantigens that bind to a site on the TCR that is outside the antigen-specific TCR binding site.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.10

26) Immunoglobulin, TCR, and MHC proteins share structural features and have evolved by duplication and selection of primordial antigen receptors.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.7

27) B cells react with antigens through TCR antigen receptors.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

28) Each immunoglobulin or TCR interacts with a single antigen, but MHC proteins can interact with more than one antigen.
Answer: TRUE
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 27.5

29) In somatic hypermutation, the mutation of immunoglobulin genes occurs at much higher rates than the mutation rates observed in other genes.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.4

30) Affinity maturation is one of the factors responsible for a stronger secondary immune response.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.4

31) In antigen binding, MHC protein binds to the T cell epitope, whereas the TCR binds the MHC motif.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.7

32) Somatic hypermutation mechanisms generate TCR diversity.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.4

33) The clonal selection theory states that each antigen-reactive B cell or T cell has a cell surface receptor for a single antigen epitope.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

34) In clonal selection, cells that have not interacted with an antigen do not proliferate.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

35) Selection against self-reactive clones results in the development of tolerance.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

36) T cells undergo immune selection FOR potential antigen-reactive cells and selection AGAINST those cells that react strongly with self-antigens.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

37) Some immature B cells are reactive to self-antigens but do not become activated even when exposed to high concentrations of self-antigens.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

38) After a B cell is activated, it no longer needs T cell interactions or cytokines to make antibody.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

27.3 Essay Questions

1) Why must the adaptive immune system develop a capacity to discriminate between foreign antigens and host antigens?
Answer: Because all macromolecules in the host are potential antigens, the host immune system must avoid recognizing host macromolecules. If the host immune system recognizes host macromolecules, antibodies or T cells would damage them.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

2) Why are the chances of an immune response in a patient given between 10μg and 1g of an immunogen (drug) higher than a patient given higher than 1g or lower than 10μg of the same drug?
Answer: Extremely high (>1g) or low (<10μg) doses may suppress the specific immune response induced by the drug by stimulating development of tolerance.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 27.1

3) What are the major antigenic barriers for tissue transplantation from one individual to another?
Answer: In different individuals, MHC proteins are not structurally identical. Different individuals have polymorphisms (subtle differences in amino acid sequences of homologous MHC proteins). Tissue transplants not matched for MHC identity are recognized by TH1 cells as nonself triggering macrophage activation and transplant destruction.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 27.5

4) Briefly describe how B cells are activated.
Answer: Antigen binds to the B cell antigen receptors. The antigen is endocysted and degraded in the B cell. Peptides from the degraded antigen are then presented on the B cell′s MHCII protein to a TH2 cell. The TH2 cell produces IL-4 and IL-5, providing the T cell help that activates the B cell. The activated B cell then differentiates into plasma cells that produce and secrete antibodies.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.8

5) What is tolerance, and why must the adaptive immune system exhibit tolerance?
Answer: Tolerance is the acquired inability to make an immune response to self-antigens. Tolerance ensures that adaptive immunity is directed to outside agents that pose genuine threats to the host and not to host proteins.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.1

6) Differentiate between T-cytotoxic (T_C) cells and T-helper (T_H) cells.
Answer: T_C cells interact with an infected cell by an MHCI protein and secrete proteins that kill the antigen-bearing infected cell. T_H cells interact with peptide-MHCII complexes on the surface of antigen-presenting cells. This interaction causes differentiation of the T_H cells into two subsets T_H1 and T_H2, which respond by proliferating and producing soluble cytokines that interact with receptors on other cells and activate them to initiate an immune response. T_H are incapable of killing infected or foreign cells. The job of T_H cells is to activate the adaptive immune response.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 27.8

7) Are superantigen reactions beneficial or harmful to the host? What is different about a superantigen reaction compared to normal antigen-host cell interactions?
Answer: Superantigen reactions are harmful to the host because too many T cells are activated, causing an excess of cytokines and inflammatory mediators to be released. Excessive inflammation, fever, and nausea occur due to superantigens. A superantigen binds to non-specific parts of the T cell receptor and MHC proteins, activating 5-25% of all T cells. Normal antigens only bind to the specific binding site of a T cell receptor that is very unique, and less than 0.01% of T cells are activated in a normal immune response to a foreign antigen.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 27.10

8) A child is stung by a bee at the beginning of the summer and has mild swelling and itching at the site of the sting. In late summer, the child is stung again, but this time rapidly develops anaphylaxis and is diagnosed with type I hypersensitivity to bee venom. If the child is allergic to bee venom, why was there no reaction after the first bee sting? What specifically happened after the first sting that led to the allergic reaction?
Answer: The child did not have a reaction after the first bee sting because there was no immunoglobulin E specific for bee venom in the child′s blood or tissues yet. After the first bee sting, mucosa-associated B cells and T_H2 cells produced cytokines that induce B cells to make IgE antibodies against the bee venom. Rather than circulating like IgG or IgM, the allergen-specific IgE antibodies bind to IgE receptors on mast cells. Mast cells are nonmotile granulocytes associated with the connective tissue adjacent to capillaries throughout the body. With any subsequent exposure to the immunizing allergen, the mast cell–bound IgE molecules bind the antigen. Cross-linking of IgEs by an antigen triggers the release of soluble allergic mediators from the mast cells, a process called degranulation. These mediators cause allergic symptoms within minutes of antigen exposure. After initial sensitization by an allergen, the allergic child will respond to each subsequent exposure to the allergen. Therefore, only after the second bee sting did the child show the signs of type I hypersensitivity or anaphylaxis. The first bee sting sensitized the child to the venom.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 27.9

9) A woman is being treated for systemic lupus erythematosus (SLE). She has an increased risk of opportunistic infections. Based on what you know about SLE and the treatment for it, propose why she has an increased risk of opportunistic infections.
Answer: SLE is a systemic autoimmune disease that involves circulating antigen-antibody complexes in the blood. The self-antigens are nucleoproteins and DNA in the blood. The immune system mistakenly produces antibodies against these self-antigens. The only way to treat SLE is to suppress the immune system to decrease the production of the self-reacting antibodies and reduce inflammation. Because the treatment suppresses the immune system, people with SLE are at greater risk of infection.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 27.9

10) You are bitten by a venomous snake at a nature center. You are rushed to the emergency room and given an antiserum. The snake handler says that he was bitten previously, but did not need treatment because he said he was exposed to venom so many times that he is immune. What is antiserum and will you have long-lasting immunity to snake venom after this? If the snake handler has truly developed some level of immunity, then how has that happened?
Answer: Antiserum is a type of artificial passive immunity where an individual is injected with pre-formed purified antibodies. In this case, the antibodies will bind to the snake venom and prevent the venom from harming the tissues. Antiserum does NOT provide long-lasting immunity. The antibodies that were injected will slowly decrease over time. Because the antibodies are given within a few hours of exposure, most people do not develop their own antibodies to snake venom after being bitten and treated. The snake handler may actually be immune to snake venom if he has been exposed multiple times without being rapidly treated. His immune system may have processed the venom as an antigen and actively created antibodies and B memory cells specific to snake venom that would provide long-lasting immunity.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 27.3

11) If you were given a new necklace and developed reddening, swelling, and itching of your skin around your neck 24 hours after putting on the necklace, what reason would you give to this phenomenon? What immune cells are involved?
Answer: This is a case of delayed-type hypersensitivity (DTH) or type IV hypersensitivity. Chemical constituents of the jewelry are covalently binding to your skin proteins and creating new antigens to elicit a DTH response known as contact dermatitis. The immune cells involved are the T_H cells. These cells produce an inflammatory response that causes localized tissue damage, pain, blistering, and itching.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 27.9

12) What role does the C domain play in cell surface proteins that interact directly with antigens?
Answer: C domains provide structural integrity for the antigen-binding molecules, attach V domains to the cytoplasmic membrane, and give each protein its characteristic shape. C domains also provide recognition sites for accessory molecules.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 27.7

13) A patient is waiting for a kidney transplant. In order for the transplant to be successful, the donated kidney (the graft) must ″match″ with the recipient. What has to ″match″ between the donor and the recipient? Why is matching tissue more difficult than matching blood types?
Answer: The immune system uses major histocompatibility complex (MHC) proteins to recognize self versus non-self cells and antigens. Each person has a unique set of MHCI proteins on all nucleated cells in the body. The MHCI proteins are composed of an alpha chain and a beta chain. The alpha chain is polymorphic, which means that there are many different variants, or alleles, in the human population. The HLA-A gene has over 2000 alleles in the human population. In addition, there are three genes that are expressed simultaneously for the MHCI alpha chain: HLA-A, HLA-B, and HLA-C. Each one has multiple alleles. Therefore, a person may have up to 6 different types of MHCI protein on the surface of every cell. The situation is similar for the MHCII proteins found on all antigen presenting cells, except there are 12 loci involved. The polymorphic and polygenic nature of the MHC proteins means that it is much more difficult to match tissues than it is to match blood types because of the huge number of possible combinations of the different alleles found in the human population. Red blood cells are non-nucleated and DO NOT express MHC proteins, thus blood type only needs to match a few proteins (A, B, and Rh antigens) in order to be compatible.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 27.6

14) Major histocompatibility complex (MHC) proteins, antibodies, and T cell receptors are all highly variable so that they can interact with a large number of different antigens. How is variability generated for each of these important antigen-binding proteins?
Answer: MHC proteins are polygenic. Three different loci for the MHCI alpha chain are expressed simultaneously, resulting in the concomitant expression of up to 6 different MHCI on the surface of nucleated cells. MHC II proteins are also polygenic and have 6 different loci. Antibodies and T cell receptors are not polygenic. Each cell expresses only one type of antibody or T cell receptor. Diversity in antibodies and T cell receptors is generated during the maturation process of the B cells (for antibodies) and T cells (for T cell receptors). This diversity is generated by somatic recombination of tandem genes, random reassortment, hypermutation, and imprecise joining of exons during the maturation of T and B cells. Each mature B cell and T cell will express one unique antibody or T cell receptor that was created and selected for during maturation.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 27.4

15) Design five amino acid peptides that could be bound by a single hypothetical class I MHC protein. The MHC proteins binds peptides that are 10 amino acids long and have two anchor residues, tyrosine and isoleucine, at positions 2 and 5.
Answer: Answers will vary because different amino acids can be used in the motif sequence. However, all five peptides should fit this pattern:
X-Y-X-X-I-X-X-X-X-X
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 27.6

16) Why is the secondary immune response stronger in regard to antibodies?
Answer: Somatic hypermutation creates B cells bearing mutated receptors after the initial exposure to an antigen. These mutated B cells then compete for available antigen. This process selects B cells with receptors having higher antigen-binding strength than the original B cell receptor. This is known as the affinity maturation process and selects for B cells that produce very high affinity antibodies for a specific antigen during secondary exposure.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 27.4

17) Given that T cell receptors (TCRs) are randomly generated during the maturation process, how are new T cells selected that can recognize self-antigens and other immune cells appropriately?
Answer: Positive selection selects for T cells that can recognize and respond appropriately to MHC-peptide complexes. Positive selection requires the interaction of new T cells in the thymus with the thymic self-antigens. Using their TCRs, some T cells bind to MHC-peptide complexes on the thymic tissue. The T cells that do not bind MHC-peptide complexes undergo apoptosis and are permanently eliminated. By contrast, those T cells that bind thymic MHC proteins receive survival signals and continue to divide and grow. Positive selection retains T cells that recognize MHC-peptide and deletes T cells that do not recognize MHC-peptide and would therefore be unable to recognize MHC-peptide outside the thymus. In the second stage of T cell maturation, T cells must be able to separate from the MHC-peptide complexes in order to continue to divide and grow. If they bind too strongly at this stage, they will not grow or finish maturation and will die. This prevents T cells that may cause immune reactions to self-antigens from maturing and entering the circulation.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 27.1